Bollywood Movies Analysis (2023–24)

Business Requirement

Business Scenario

A movie production company wants to understand the factors that contribute to the success of Bollywood movies. By analyzing historical movie data, the company aims to identify popular genres, highly rated directors, successful actors, and audience preferences. These insights will help the production team make better decisions while selecting movie genres, directors, and cast members for future films.

As a **Junior Data Analyst**, your responsibility is to analyze the Bollywood movie dataset using Python and Pandas. You will perform data cleaning, exploratory data analysis (EDA), and visualization to answer important business questions and provide recommendations to the management team.

Project Objective

Analyze the Bollywood movie dataset to:

- Understand trends in movie production over the years.
- Identify the most popular genres.
- Analyze movie ratings and audience votes.
- Evaluate the performance of directors and actors.
- Generate meaningful business insights and recommendations.

Dataset Description

The dataset contains information about **Indian movies** including their release year, duration, genre, IMDb rating, audience votes, directors, and lead actors.

Dataset Features

Column	Description
Name	Movie Name
Year	Release Year
Duration	Movie Duration (Minutes)
Genre	Movie Genre
Rating	IMDb Rating
Votes	Number of Audience Votes
Director	Movie Director
Actor 1	Lead Actor
Actor 2	Supporting Actor
Actor 3	Supporting Actor

Tasks to Perform

Task 1 – Data Loading

- Import the required libraries.
- Load the dataset using Pandas.
- Display the first five records.
- Display the last five records.
- Find the number of rows and columns.
- Display all column names.
- Check the data types of each column.

Task 2 – Data Exploration

Perform the following analysis:

- Check for missing values.
- Check duplicate records.
- Display summary statistics.
- Find the number of unique directors.
- Find the number of unique genres.
- Find the number of unique actors (Actor 1).

Task 3 – Data Cleaning

Perform the following operations:

- Remove duplicate records (if any).
- Handle missing values.
- Ensure **Year**, **Duration**, **Rating**, and **Votes** have appropriate data types.
- Remove unnecessary spaces from text columns.

Business Questions

Answer the following questions using Python and Pandas.

Movie Analysis

- How many movies were released each year?
- Which year had the highest number of movie releases?
- What is the average IMDb rating of all movies?

Genre Analysis

- Which genre has the highest number of movies?
- Which genres have the highest average IMDb ratings?
- Compare the ratings across different genres.

Director Analysis

- Which directors have directed the highest number of movies?
- Which directors have the highest average movie ratings?
- Display the Top 10 directors based on IMDb ratings.

Actor Analysis

- Which lead actors (Actor 1) have appeared in the highest number of movies?
- Which lead actors have the highest average IMDb ratings?
- Display the Top 10 lead actors based on movie count.

Audience Analysis

- Which movies received the highest number of audience votes?
- Which movies have the highest IMDb ratings?
- Is there any relationship between audience votes and movie ratings?

Duration Analysis

- What is the average movie duration?
- Which are the Top 10 longest movies?
- Which genre has the highest average movie duration?

Data Visualization

Create **at least five charts** using Matplotlib.

Mandatory Charts

- Movies Released by Year (Line Chart)
- Top 10 Movie Genres (Bar Chart)
- Top 10 Directors by Number of Movies (Bar Chart)
- IMDb Rating Distribution (Histogram)
- Top 10 Movies by Audience Votes (Horizontal Bar Chart)

Optional Chart

- Top 10 Lead Actors by Number of Movies (Bar Chart)

Business Insights

Write **at least 5 business insights** based on your analysis.

Examples:

- Which genres are produced the most?
- Which directors consistently receive higher IMDb ratings?
- Which actors frequently appear in successful movies?
- Which movies received the highest audience engagement?
- Are highly rated movies also receiving more audience votes?

Business Recommendations

Provide **at least 3 recommendations** for the movie production company.

Example recommendations:

- Invest more in genres that consistently receive higher IMDb ratings.
- Collaborate with directors who have a strong track record of successful movies.
- Cast actors who frequently appear in highly rated films.
- Consider audience preferences while selecting future movie projects.

Submission Requirements

Students must submit:

- Python Notebook (.ipynb)
- Cleaned Dataset (if modified)
- Report
- 3 Business Recommendations

- 5-minute PowerPoint Presentation

Evaluation Criteria

Criteria	Marks
Data Loading & Exploration	15
Data Cleaning	15
Data Analysis	30
Visualizations	20
Business Insights	10
Recommendations	10
Total	**100**